

Algebraic Manipulation

Algebraic manipulation involves applying algebraic operations, such as addition, subtraction, multiplication, and division, to reorganize equations. This process often aims to isolate a particular variable, making it easier to solve equations or understand the relationship between different variables.

Special Cases

I) Make r the subject of the expression $v = \frac{2}{3}\pi r^3$. To make r the subject of the expression $v = \frac{2}{3}\pi r^3$, is necessary to isolate r on one side of the equation. Here's how to do it:

Given: $v = \frac{2}{3}\pi r^3$

Divide both sides by $\frac{2}{3}\pi$ to isolate r^3 :

$$\begin{aligned}\frac{v}{\frac{2}{3}\pi} &= \frac{\frac{2}{3}\pi r^3}{\frac{2}{3}\pi} \\ \frac{v}{\frac{2}{3}\pi} &= \frac{\cancel{\frac{2}{3}}\pi r^3}{\cancel{\frac{2}{3}}\pi} \\ \frac{3v}{2\pi} &= r^3\end{aligned}$$

Now, to solve for r , take the cube root of both sides:

$$\begin{aligned}\sqrt[3]{\frac{3v}{2\pi}} &= \sqrt[3]{r^3} \\ \sqrt[3]{\frac{3v}{2\pi}} &= \sqrt[3]{r^3} \\ r &= \sqrt[3]{\frac{3v}{2\pi}}\end{aligned}$$

So, r is the subject of the expression $v = \frac{2}{3}\pi r^3$, and it can be written as $r = \sqrt[3]{\frac{3v}{2\pi}}$.

II) Make b the subject of the expression $\frac{p}{q} = \sqrt{\frac{a+2b}{a-2b}}$. To make b the subject of the expression $\frac{p}{q} = \sqrt{\frac{a+2b}{a-2b}}$, is necessary to isolate b on one side of the equation. Here's how to do it:

Given: $\frac{p}{q} = \sqrt{\frac{a+2b}{a-2b}}$

Square both sides to remove the square root:

$$\left(\frac{p}{q}\right)^2 = \left(\sqrt{\frac{a+2b}{a-2b}}\right)^2$$

$$\left(\frac{p}{q}\right)^2 = \frac{a+2b}{a-2b}$$

$$\left(\frac{p}{q}\right)^2 = \frac{a+2b}{a-2b}$$

$$\frac{p^2}{q^2} = \frac{a+2b}{a-2b}$$

Multiply both sides by $a-2b$ and q^2 to clear the fraction (or just multiply crosswise):

$$\frac{p^2}{q^2} \cdot (a-2b) \cdot q^2 = \frac{a+2b}{a-2b} \cdot (a-2b) \cdot q^2$$

$$\frac{p^2}{q^2} \cdot (a-2b) \cdot \cancel{q^2} = \frac{a+2b}{\cancel{a-2b}} \cdot \cancel{a-2b} \cdot q^2$$

$$p^2 \cdot (a-2b) = q^2 \cdot (a+2b)$$

Expand the multiplication:

$$p^2 \cdot a - 2 \cdot p^2 \cdot b = q^2 \cdot a + 2 \cdot q^2 \cdot b$$

Rearrange the terms adding $2 \cdot p^2 \cdot b$ and subtracting $q^2 \cdot a$ in both sides to group those with b :

$$p^2 \cdot a - 2 \cdot p^2 \cdot b - q^2 \cdot a + 2 \cdot p^2 \cdot b = q^2 \cdot a + 2 \cdot q^2 \cdot b - q^2 \cdot a + 2 \cdot p^2 \cdot b$$

$$p^2 \cdot a - \cancel{2 \cdot p^2 \cdot b} - q^2 \cdot a + \cancel{2 \cdot p^2 \cdot b} = \cancel{q^2 \cdot a} + 2 \cdot q^2 \cdot b - \cancel{q^2 \cdot a} + 2 \cdot p^2 \cdot b$$

$$p^2 \cdot a - q^2 \cdot a = 2 \cdot q^2 \cdot b + 2 \cdot p^2 \cdot b$$

Now is possible to factor out b and a :

$$(p^2 - q^2) \cdot a = b \cdot 2 \cdot (p^2 + q^2)$$

To isolate b , divide both sides by $2 \cdot (p^2 + q^2)$:

$$\frac{(p^2 - q^2) \cdot a}{2 \cdot (p^2 + q^2)} = b \cdot \frac{2(p^2 + q^2)}{2 \cdot (p^2 + q^2)}$$

$$\frac{(p^2 - q^2) \cdot a}{2 \cdot (p^2 + q^2)} = b \cdot \frac{\cancel{2(p^2 + q^2)}}{\cancel{2 \cdot (p^2 + q^2)}}$$

$$b = \frac{a \cdot (p^2 - q^2)}{2 \cdot (p^2 + q^2)}$$

So, b the subject of the expression $\frac{p}{a} = \sqrt{\frac{a+2b}{a-2b}}$, and it can be written as $b = \frac{a \cdot (p^2 - q^2)}{2 \cdot (p^2 + q^2)}$

III) Solve for t : $\frac{15}{3t^2} = \frac{2}{5}$

To solve for t in the equation $\frac{15}{3t^2} = \frac{2}{5}$, is necessary:

Given that there is a proportion, is possible to cross-multiply to eliminate the fractions:

$$15 \cdot 5 = 2 \cdot (3t^2)$$

$$75 = 6t^2$$

Now, isolate t^2 by dividing both sides of the equation by 6:

$$t^2 = \frac{75}{6}$$

$$t^2 = 12.5$$

Since you're interested in solving for t , take the square root of both sides to isolate t :

$$t = \pm\sqrt{12.5}$$

$$t = \pm 3.54$$

Thus, the solution to the given expression is:

$$t = \pm 3.54$$